🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 5 · STANDARD 6.GR.1

Determine the Area of Triangles

Lesson 5-2 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

SKY HARBOR MISSION

The Triangular Skylight

You are the lead architect at Sky Harbor Studio. The city has hired your firm to design a glass skylight made of triangular panels for the new transit hub. Before the glass can be cut, you must master the area of triangles — every wrong measurement wastes expensive glass.

My Notes Level 1 Support

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can find the area of a triangle using the formula A = ½ × base × height.

Language Objective: I can explain my steps using the words base, height, area, and perpendicular.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Area of a Triangle | Base | Height | Area | Perpendicular | Composite figure | Formula | Parallel

👆 *Tap any word to see what it means and a picture.*

1. The space inside a triangle, found with $A = \frac{1}{2} \times b \times h$, is the ________.

2. The side of a triangle you measure the height from at a 90° angle is the ________.

3. The perpendicular distance from the base to the opposite vertex is the ________.

4. The amount of flat space inside a triangle is its ________.

5. Two lines that meet at a 90° angle are ________.

6 A shape made of two or more basic shapes combined is a

________________ .

7 A math rule written with symbols, like $A = \frac{1}{2} \times b \times h$, is a

________________ .

8 Two sides that stay the same distance apart and never meet are

________________ .

Write About the Math

How many cans of paint does the homeowner need?

¿Cuántas latas de pintura necesita el dueño de la casa?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Area of a Triangle** — *Área de un triángulo*
- ☐ **Base** — *Base*
- ☐ **Height** — *Altura*
- ☐ **Area** — *Área*
- ☐ **Perpendicular** — *Perpendicular*

Model: *The height is the perpendicular distance from the base to the top corner.* — Students identify the height as the perpendicular (straight-up, 90-degree) distance from the base to the opposite vertex, not the slanted side.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The wall's area is ____ sq ft because A = 1/2 × ____ × ____. She needs ____ cans because ____ ÷ 50 = ____.**
- **My answer is ____ because ____.**
  Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **My evidence is ____, so ____.**
  Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Area of a Triangle
2. **Notes 2:** Base
3. **Notes 3:** Height
4. **Notes 4:** Area
5. **Notes 5:** Perpendicular
6. **Notes 6:** Composite figure
7. **Notes 7:** Formula
8. **Notes 8:** Parallel
9. **Watch for this mistake:** *A common mistake in Area of Triangles is forgetting the ½ — students multiply base × height like a rectangle and stop there, or they use the triangle's slanted side instead of the perpendicular height that's already given. Always check: did I take half, and did I use the straight-up height, not the slanted side?*
10. **Try It 1:** A. 14 sq ft *— A = ½ × base × height = ½ × 7 × 4 = ½ × 28 = 14 sq ft. Multiply base × height first (28), then take half (14).*
11. **Try It 2:** A. 33 sq ft *— A = ½ × base × height = ½ × 11 × 6 = ½ × 66 = 33 sq ft.*